INCH-POUND

MS27247L
22 July 1992
SUPERSEDING
MS27247K
10 February 1989

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2PDT, TYPE I, HERMETICALLY SEALED

Inactive for new design after 5 June 1987.
No superseding standard. For new design,
use MS27401.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.

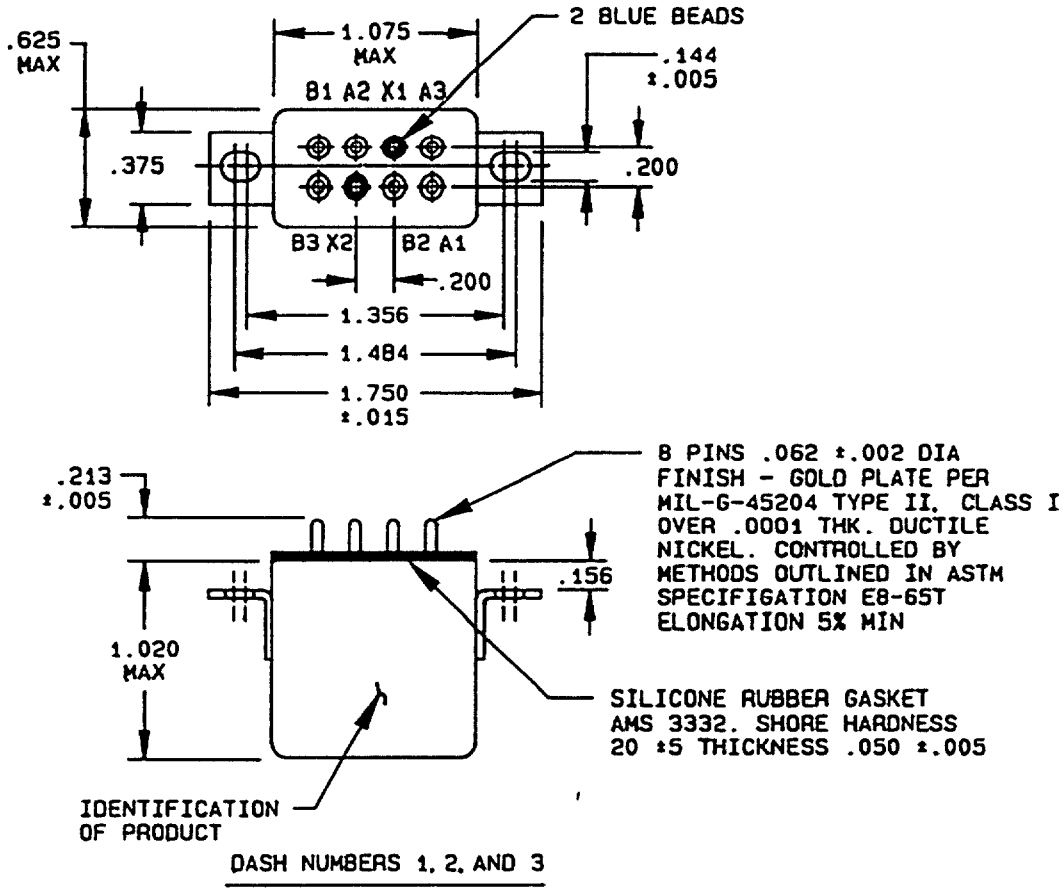
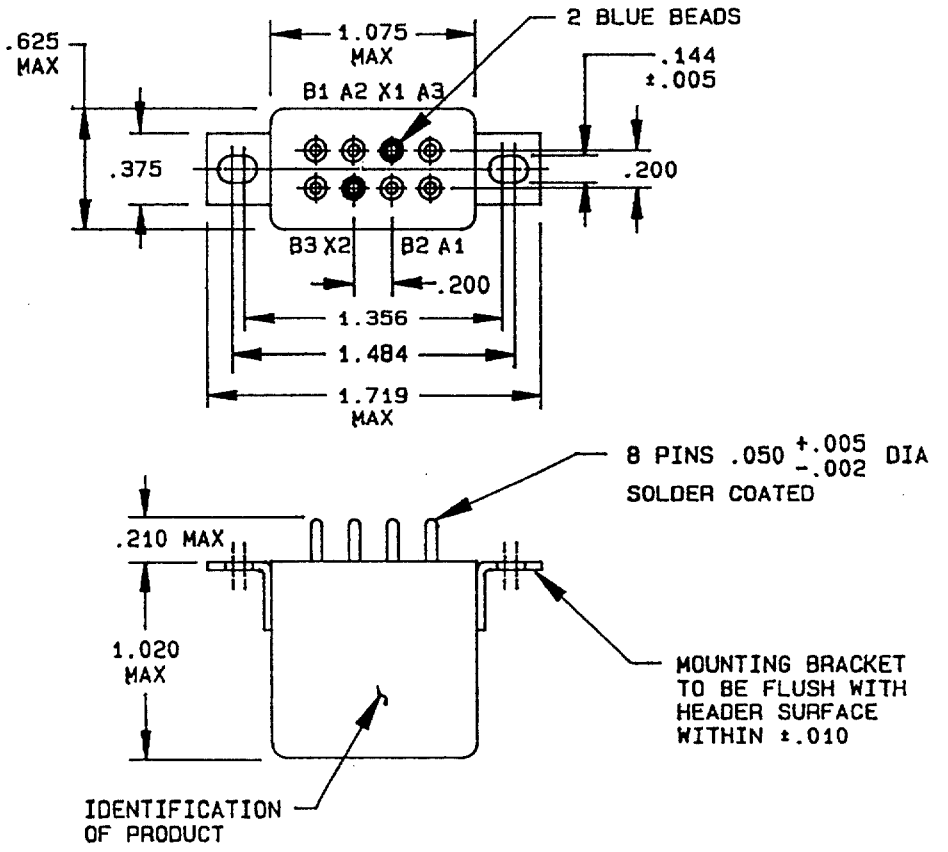


FIGURE 1. Dimensions and configurations.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

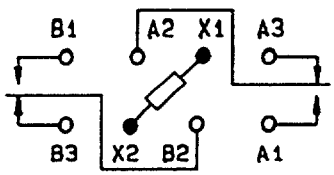
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DASH NUMBERS 4 AND 5

FIGURE 1. Dimensions and configurations - Continued.

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CIRCUIT DIAGRAM

Inches	mm	Inches	mm
.0001	0.025	.210	5.33
.002	0.05	.213	5.41
.005	0.13	.375	9.52
.010	0.25	.515	13.08
.015	0.38	1.000	25.40
.050	1.27	1.015	25.78
.062	1.57	1.356	34.44
.144	3.66	1.484	37.69
.156	3.96	1.719	43.66
.200	5.08	1.750	44.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay headers provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

ENVIRONMENTAL CHARACTERISTICS:

Temperature range: -70°C and +125°C.

Maximum altitude rating: 80,000 feet.

Shock G-level: 50 g's.

Duration: 11 ms.

Max duration contact opening: 10 µs.

Vibration - sinusoidal:

G-level: 20 g's.

Frequency range: 10 - 2,000 Hz.

Acceleration: 15 g's.

ELECTRICAL CHARACTERISTICS (see tables I, II, and III).

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric strength (sea level):

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Aux contacts		
All other points	1,250 V rms	1,000 V rms

Dielectric strength (altitude): 80,000 feet.

	Initial	After life tests
Coil to case		350 V rms
Aux contacts		
All other points		350 V rms

Max contact drop initial: 0.150 volt.

After life test: 0.175 volt.

Overload current: 20 amperes, 28 V dc.

Rupture current: 40 amperes, 115 V ac, 400 Hz.

Duty rating: Continuous.

QUALITY CONFORMANCE INSPECTION:

Performance of groups B and C tests are not applicable.

Group A acceptance reports shall be submitted to the qualifying activity on a yearly basis in order to retain qualification for this military specification sheet.

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TABLE I. Dash numbers and characteristics.

Dash number MS27247-	Type	Coil	Terminal type	115/200 V ac 3 phase contact rating	Mounting or mating sockets	Max weight in pounds
1	I	dc	Plug-in	N/A	N/A	.08
2	I	dc	Plug-in	Applicable	N/A	.08
3	IER	dc	Plug-in	Applicable	N/A	.08
4	I	dc	Printed wire (PW)	N/A	N/A	.08
5	I	dc	Printed wire (PW)	Applicable	N/A	.08

TABLE II. Operating characteristics.

MS part number	Coil data											Time - milliseconds max					
		Rated			Max		Max pick-up voltage			Hold voltage 2/	Drop out voltage 2/	Operate 3/	Release 4/	Contact bounce			
														Main		Aux	
	Coil	Volts 1/	Freq. Hz	Res Ω ±10%	Volts	Ampere	Normal 2/	High test temp	Cont current test					NO	NC	NO	NC
MS27247-1	dc	28	dc	300	29	.120	18	18	19.8	7	1	10	10	2	5		
MS27247-2	dc	28	dc	300	29	.120	18	18	19.8	7	1	10	10	2	5		
MS27247-3	dc	28	dc	300	29	.120	18	18	19.8	7	1	10	10	2	5		
MS27247-4	dc	28	dc	300	29	.120	18	18	19.8	7	1	10	10	2	5		
MS27247-5	dc	28	dc	300	29	.120	18	18	19.8	7	1	10	10	2	5		

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With rated coil voltage.
4/ From rated coil voltage.

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TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3 1/	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate footnotes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10			10				10				
Inductive	100	4	4			4				4				
Inductive 2/ (-5 only)														
Motor	100	3	3			3				3				
Lamp	100	2	2			2				2				
Transfer load														3/
Mechanical life reduced current	400	2.5	2.5											
Intermediate current														4/

1/ 115/200 V ac for 60 Hz ratings, absence of value indicates relay is not rated for 3 phase applications.
2/ Additional rating for -5 only: 2 ampere inductive at 460 V ac, 400 Hz.
3/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.
4/ Minimum current ratings 0.1 ampere resistive, 0.3 ampere inductive, and 0.5 ampere resistive are applicable in accordance with acquisition specification except monitored to a 3 ohm contact resistance.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Navy - EC
Air Force - 85

Review activities:
Navy - AS
Air Force - 99

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0886)

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DEPARTMENT OF THE AIR FORCE
2750 LOG/ES
STANDARDIZATION
CUSTOMER SATISFACTION INQUIRY

TELEPHONE - COM 513-296-5517 DSN 986-5517
FAX - COM 513-296-8783 DSN 986-8783

VERY DISSATISFIED	DISSATISFIED	SOMEWHAT DISSATISFIED	SOMEWHAT SATISFIED	SATISFIED	VERY SATISFIED	NOT APPLICABLE
1	2	3	4	5	6	7

(PLEASE ENTER THE VALUE THAT YOU FEEL IS APPROPRIATE)

I. STAFF

1. PROVIDERS QUALIFICATIONS/COMPETENCE ☐

2. PROVIDERS PROFESSIONALISM ☐

	TIMELINESS	QUALITY
3. PROVIDERS RESPONSE TO YOUR INITIAL REQUEST	☐	☐

II. PRODUCT

	TIMELINESS	QUALITY
1. INITIAL DRAFT	☐	☐
2. REVISION/AMENDMENT SATISFIED YOUR REQUIREMENT	☐	☐

III. COMMENTS/SUGGESTIONS

(CONSTRUCTIVE COMMENTS OR SUGGESTIONS APPRECIATED)

MS27247L

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